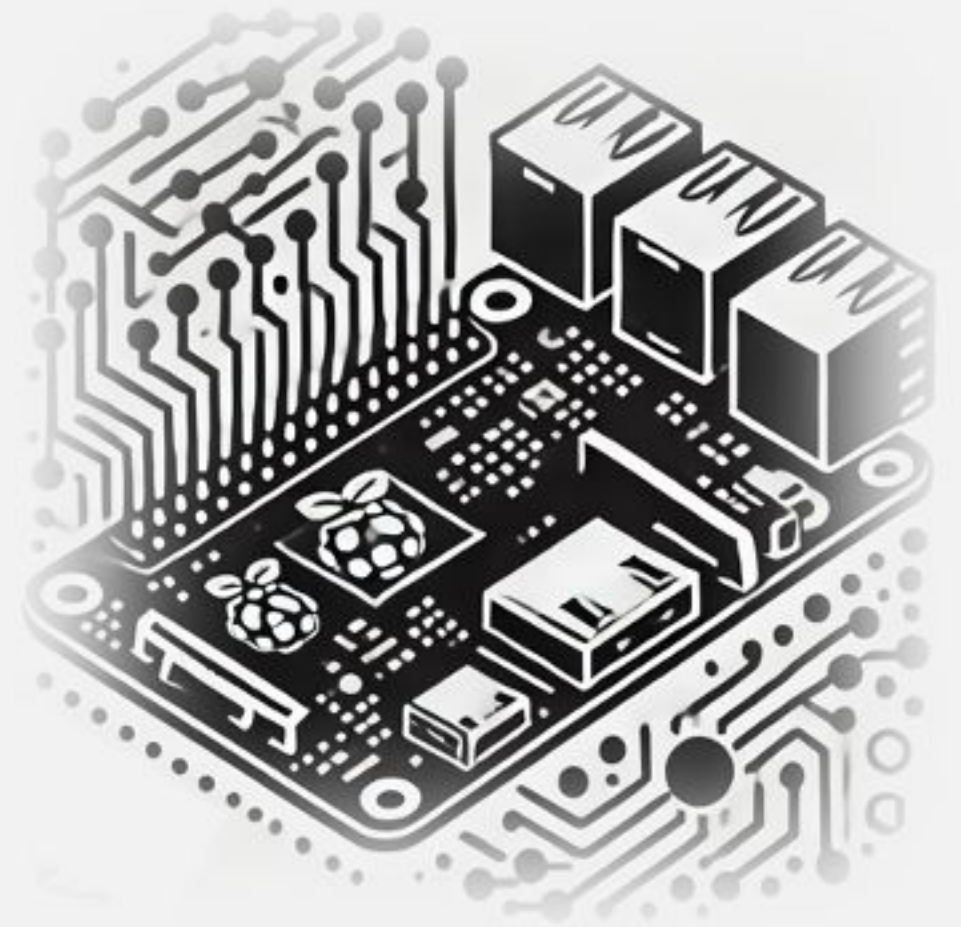


IESTI05 – Edge AI

Machine Learning System Engineering

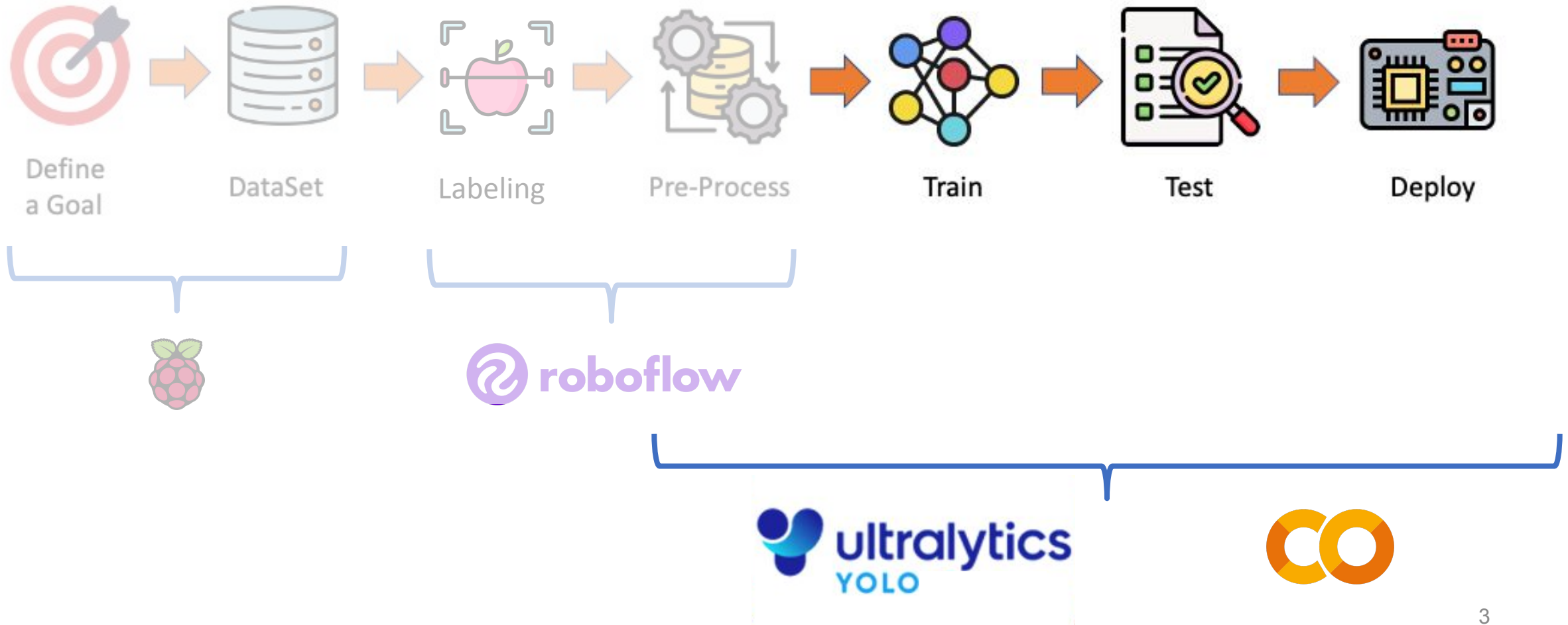
12. Object Detection Project: Train & Deploy – YOLO



The Goal



Machine Learning Workflow



Data Aquisition



Universe Search 500,000+ Open Source Computer Vision Projects...

Box versus Whe... Object Detection

Versions

v8 2025-09-01 2:56pm
Generated on Sep 1, 2025

Download Dataset

Download

Image and Annotation Format

YOLOv11

TXT annotations and YAML config used with YOLOv11.

Download Options

☐ Download zip to computer
Downloads all images, annotations, and classes.

☒ Show download code
Custom train this dataset using the provided code snippet in a notebook.

Cancel Continue

248 Total Images View All Images →

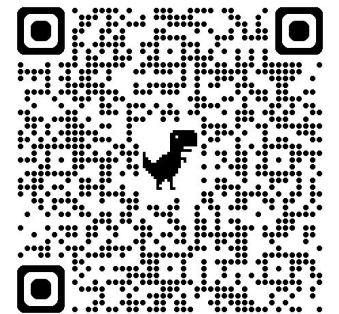
Dataset Split

Set	Percentage
TRAIN SET	91%
VALID SET	5%
TEST SET	4%

5

Ultralytics Training:
[yolo11_box_vs_wheel.ipynb](#)

[\[OPEN COLAB\]](#)



yolo11_box_vs_wheel.ipynb

https://colab.research.google.com/drive/1tvVtXFoKlZBT1RFbpq8C9ODjygBBuVuS#scrollTo=bSq6KopwNxnF

CO

yolo11_box_vs_wheel.ipynb

☆

☁

File Edit View Insert Runtime Tools Help

Q Commands + Code + Text ▶ Run all

✓ RAM Disk

Share Gemini

Files

..

datasets

Box-versus-Wheel-auto-datase..

test

images

labels

train

images

labels

labels.cache

valid

README.dataset.txt

README roboflow.txt

data.yaml

Disk 73.16 GB available

Dataset

[5] 1 !mkdir {HOME}/datasets
2 %cd {HOME}/datasets
3
4 !pip install roboflow --quiet
5
6 from roboflow import Roboflow
7 rf = Roboflow(api_key="8NEOn2tX76EJbk9hQTb4")
8 project = rf.workspace("marcelo-rovai-riila").project("box-versus-wheel-auto-dataset")
9 version = project.version(8)
10 dataset = version.download("yolov11")
11

PAST

/content/datasets
88.6/88.6 kB 5.2 MB/s eta 0:00:00
66.8/66.8 kB 6.8 MB/s eta 0:00:00
49.9/49.9 MB 20.3 MB/s eta 0:00:00
1.4/1.4 MB 75.1 MB/s eta 0:00:00
4.2/4.2 MB 108.1 MB/s eta 0:00:00
loading Roboflow workspace...
loading Roboflow project...
Downloading Dataset Version Zip in Box-versus-Wheel-auto-dataset-8 to yolov11:: 100%|██████████| 4855/4855 [00:
Extracting Dataset Version Zip to Box-versus-Wheel-auto-dataset-8 in yolov11:: 100%|██████████| 508/508 [00:00:

Variables Terminal

2:55 PM T4 (Python 3)

yolo11_box_vs_wheel.ipynb

https://colab.research.google.com/drive/1tvVtXFoKlZBT1RFbpq8C9ODjygBBuVuS#scrollTo=cK5WknXSVOoN

File Edit View Insert Runtime Tools Help

Commands Code Text Run all

Files

- datasets
- gdrive
- runs
 - detect
 - predict
 - train
 - weights
 - best.pt
 - last.pt
 - BoxF1_curve.png
 - BoxPR_curve.png
 - BoxP_curve.png
 - BoxR_curve.png

30s

Export the Training Result to the Gdrive

```
[15] 1 from google.colab import drive
      2 drive.mount('/content/gdrive')
```

Mounted at /content/gdrive

```
[ ] 1 /content/gdrive/MyDrive/10_UNIFEI/IESTI05-Box-Wheel-2025
```

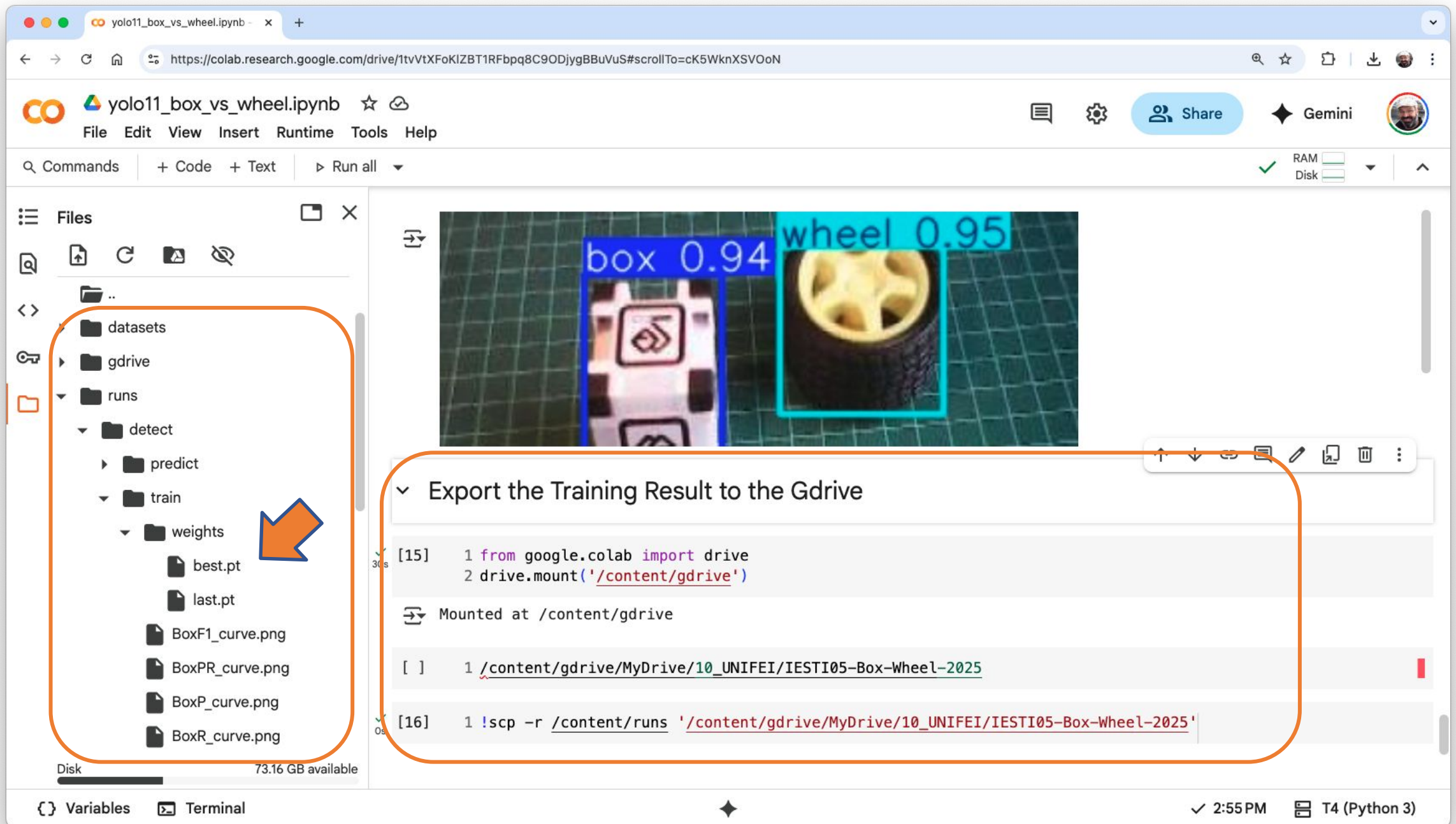
0s

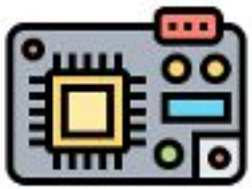
```
[16] 1 !scp -r /content/runs '/content/gdrive/MyDrive/10_UNIFEI/IESTI05-Box-Wheel-2025'
```

Disk 73.16 GB available

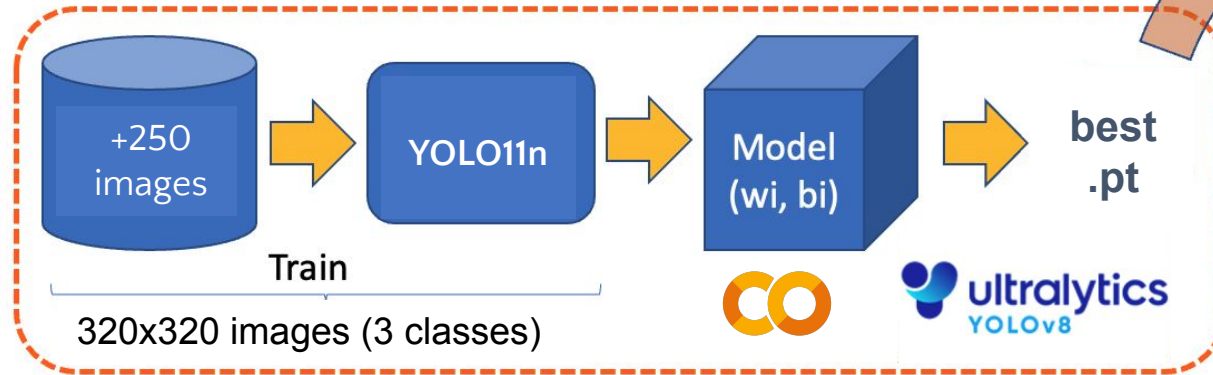
Variables Terminal

2:55 PM T4 (Python 3)

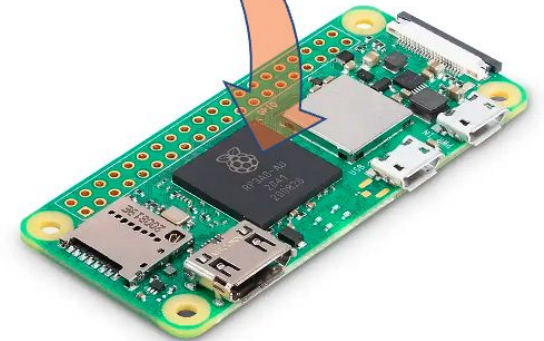




Ultralytics YOLO11n



5.5 MB



Raspberry Pi Inference:

YOLO Model Predictions with Ultralytics.ipynb



Questions?



Prof. Marcelo J. Rovai

rovai@unifei.edu.br



UNIFEI